MORAURA

TEAM PROJECT PROPOSAL

Smart soil condition monitoring

An ESP32-powered system for healthier plants and more precise care.

Why MorAura?

Monitoring plants is a tedious labour intensive task, and strict monitoring can reduce the likelihood of plants dying out and even improve the yield of crops. Our system is designed to monitor plant conditions to make sure they're within the ideal ranges.

01

Accurately measure plant conditions.

02

Reduce guesswork in plant care.

03

Reduce the need for constant human monitoring.

System architecture

SENSORS USED

- Soil moisture
- Light intensity
- Temperature
- Humidity

ESP32

Simple mobile app

Bluetooth Low Energy

Adjust monitoring parameter ranges

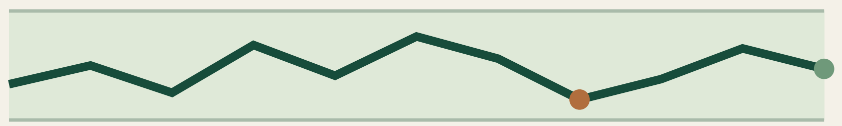
Monitoring profile

Illustrative monitoring profile — values are examples for the prototype.

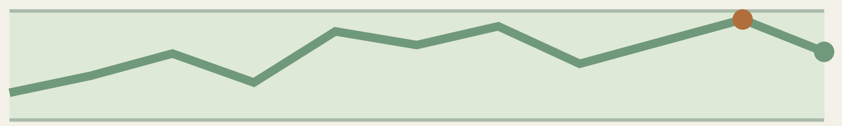
TARGET RANGES

Moisture	**40 — 65%**
Light	**1k — 10k lux**
Temperature	**20 — 28°C**
Humidity	**60 — 80%**

MOISTURE LOG · %



TEMPERATURE LOG · °C



Alert when a reading leaves its target range. **OUT OF RANGE**

Build the prototype. Test it in the real world.

01

Research the problem and sensors.

02

Assemble a prototype.

03

Visit plantations, survey, and collect feedback.

04

Calibrate the ranges and improve the prototype based on feedback.

05

Test at real plantation sites.

Visit plantation sites to observe, survey, and collect information and feedback about the system, ensuring it is practical for real-world scenarios.